Ilia Mirzaali

Software Engineer

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Dutch student residence permit — 16 h/week during term, full-time June–August.

Profile

Software engineer and second-year Computer Science student at the University of Twente (BSc, expected Jul. 2028, weighted average 8.42/10). Experience across frontend, backend, full-stack development, systems programming and security. Contribute to the Quicksports startup's Node.js and Vue 3 platform, including APIs, dashboards and cloud deployments. Run a penetration testing consultancy and build systems projects including a Redis-compatible C++ server and an eBPF kernel telemetry instrument. BAPC competitor, qualified for NWERC.

Experience

Ramsy (ramsy.eu) Enschede, Netherlands
Founder & Engineer *Mar. 2026 – Present*

- Founded a KvK-registered security consultancy serving Dutch businesses and have delivered 20+ engagements; run it end to end, from client scoping and pricing to delivery, reporting, and retesting.
- Designed and built the internal tooling the business runs on: automated asset and subdomain discovery, scan orchestration, and templated report generation.
- Findings include an unauthenticated DICOM/PACS archive exposing roughly 145,000 medical imaging studies, and a client host already occupied by a cryptomining botnet via an exposed Redis instance — traced to its command-and-control and disclosed the same day.

Quicksports B.V. Enschede, Netherlands
Software Engineer (startup contributor) *Oct. 2025 – Present*

- Lead and mentor a team of developers and interns, owning onboarding, code review, and task delegation while keeping delivery velocity and code quality steady.
- Architected a real-time event-driven messaging microservice on Node.js, Socket.IO, and Redis Pub/Sub with MongoDB caching, delivering near-instant updates to community members.
- Led migration of 5 microservices from Heroku to Google Cloud Run with Cloud Build CI/CD, containerised deployments, and cost-optimised autoscaling, cutting infrastructure spend by 30%.
- Own the Organizer Dashboard end to end: gather requirements directly from customers, iterate on their feedback, and ship full-stack features in Vue 3, Vuetify, and Node.js.

Full Stack Developer (startup contributor) *May 2025 – Oct. 2025*

- Shipped the Communities feature and a rebuilt admin dashboard, contributing to a 20% increase in the user base.
- Built and maintained REST APIs, AI pipelines, and internal tools on Node.js, Express, MongoDB, and Redis; worked across the full cycle including testing, CI/CD, and deployment.

DroneTeam Twente Enschede, Netherlands
Chief of Dronobox *Apr. 2026 – Present*

- Built and maintain droneteamtwente.nl, the team's public website.

Dronobox Engineer *Sep. 2025 – Apr. 2026*

- Co-organized drone workshops, mentored students in building and flying drones, and maintained and repaired the team's drone fleet.

Projects

Cedis — Redis-Compatible Server in C++ *C++17, Boost.Asio, CMake, GitHub Actions*

- Asynchronous, non-blocking in-memory key-value server built from scratch, implementing the RESP wire protocol with a modular command parser and dispatcher.
- Cross-platform CMake build with GitHub Actions CI across Windows, macOS, and Linux.

Khor — Kernel Telemetry Instrument *C++, eBPF, React, TypeScript, Vite*

- Real-time Linux daemon collecting kernel tracepoint data (process exec, network, scheduler, disk I/O, interrupts) through eBPF and normalising it into a deterministic control signal.
- Custom C++ polyphonic synthesiser with multi-channel MIDI and OSC output, plus a React and TypeScript dashboard communicating with the daemon over a low-latency transport.

ChokerJoker — Quarto Game AI *Java, alpha-beta search*

- Winner of the University of Twente Module 2 algorithms competition; combines a heuristic midgame that minimises the opponent's safe moves with an exact endgame solver.
- Made exhaustive search tractable with bitboards, Zobrist hashing, 32-way symmetry reduction, and tuned move ordering for alpha-beta pruning.

Tartarus — Malware Analysis Sandbox *Rust, KVM/QEMU, eBPF*

- Learning project exploring isolated malware execution with KVM/QEMU and eBPF-based behavioural telemetry. In active development.

Education

University of Twente Enschede, Netherlands
BSc Computer Science — expected Jul. 2028 *Sep. 2025 – Present*

- Weighted average 8.42/10.

Twente Pathway College Enschede, Netherlands
Foundation Year, General Engineering *Sep. 2024 – Jun. 2025*

- Grade: 8.9/10.

NODET (Sampad), National Organisation for Development of Exceptional Talents Gorgan, Iran
High School Diploma, Mathematics & Physics *Graduated Jun. 2023*

- Admission by nationwide competitive examination to Iran's selective school system.

Achievements & Activities

- 1st place of 13 teams, Amsterdam Fixathon 2026 (Stichting Single SuperMom challenge) — built an AI-driven eligibility engine turning Dutch benefits regulation into machine-checkable rules, using hybrid search and a self-correcting scraper.
- 2nd place among University of Twente teams, Benelux Algorithmic Programming Contest (BAPC) 2025 — qualified for the Northwestern European Regional Contest (NWERC).
- 3rd place, Twente Algorithmic Programming Contest (TAPC) 2025.
- Co-founder, C3ntaur — student CTF team ranked among the top teams in the Netherlands; 48th internationally at Nullcon Goa HackIM 2026 (Sep. 2025 – Present).

Skills

Languages: C, C++17, Rust, Python, Java, TypeScript, JavaScript, PHP, SQL, Bash, HTML/CSS
Backend & web: Node.js, Express, Vue 3, Vuetify, React, Next.js, Socket.IO, Flask, FastAPI, REST API design, WebSockets
Data & messaging: MongoDB, Redis (incl. Pub/Sub), SQL
Systems: Linux, eBPF, KVM/QEMU, Boost.Asio, SDL2, CMake, vcpkg, OpenSSL, systemd, concurrency and async I/O
Infrastructure: Docker, Google Cloud Run, Cloud Build, AWS (EC2, S3, Lambda), Nginx, CI/CD (GitHub Actions), Git
Security: OWASP Top 10, OWASP WSTG, threat modelling, Burp Suite, secure code review
Spoken languages: Persian (native), Turkmen (native), English (C1), Turkish (B2), Dutch (A1, actively learning)